# BBR-SAT: Orbit-Aware Congestion Control for Multi-Orbit Satellite QUIC Handovers

Rajat Bhambani
*Independent Research Contribution*
rajatbhambani@gmail.com

***Abstract*—Enterprise and maritime operators increasingly combine LEO, MEO, and GEO satellite links into unified multi-WAN architectures using SD-WAN devices that preserve transport sessions across WAN switches. When traffic fails over between orbit classes — whether via an SD-WAN policy decision today or a single-terminal beam switch in the near future — the surviving TCP or QUIC connection experiences an abrupt, step-change shift in path characteristics that existing congestion control algorithms (CCAs) handle poorly: BBRv3 stalls for the duration of its two-cycle bandwidth filter, and simple heuristics such as congestion-window freeze or send-pause succeed only at a single, carefully tuned lead time. We present BBR-SAT, a minimal extension of BBRv3 that equips the sender with an orbit parameter table seeded from ephemeris data and a two-phase handover protocol: a PREDICTED signal initiates a proactive queue drain via ProbeBW_DOWN, and a CONFIRMED signal adaptively resets the BDP context from the orbit table for downward transitions (GEO target) while acting as a no-op for upward transitions that self-correct naturally. Implemented in picoquic and evaluated in a zero-loss shared-link simulator across all six pairwise LEO/MEO/GEO transitions at three handover times and six advance lead times, BBR-SAT converges to 90% of new-orbit capacity in as little as 1.5 s with a 2-second advance signal (4.5 s at zero lead time) for the critical LEO→GEO transition under a realistic 1×BDP droptail buffer cap — the only zero-loss algorithm to converge; CUBIC converges in 5.5 s via packet loss, while all BBR baselines fail (N/C, $\leq 71$ % utilisation). In the uncapped large-buffer scenario, BBR-SAT reduces peak queue by $44\times$ over vanilla BBRv3. Fairness analysis shows that BBR-SAT co-exists equitably with competing BBRv3 flows (Jain $J = 0.994$) and neither worsens nor repairs the pre-existing BBRv3 deference to CUBIC.**

***Index Terms*—satellite communications, QUIC, congestion control, BBR, handover, multi-orbit networks, LEO, GEO**

## I. Introduction

The deployment of large LEO constellations (Starlink, OneWeb, Amazon Kuiper) [1], [2] alongside legacy MEO (O3b, O3b mPOWER) and GEO (ViaSat-3) infrastructure has created a heterogeneous satellite ecosystem. Today, enterprise and maritime operators routinely combine terminals from multiple constellations into unified multi-WAN architectures using SD-WAN devices (Peplink, Cisco, Fortinet) that bond LEO, MEO, and GEO links for resilience and load balancing. These SD-WAN platforms preserve transport-layer sessions across WAN switches via overlay tunnels or split-TCP proxies: when the SD-WAN fails traffic over from a LEO link to a GEO link, the TCP or QUIC connection's 5-tuple survives, but the congestion control algorithm (CCA) is abruptly exposed to a path with fundamentally different characteristics — without any advance notification. In the near future, single multi-beam terminals capable of accessing multiple orbital planes will further increase the frequency of such transitions. Whether triggered by an SD-WAN policy decision or a single-terminal beam switch, the transport-layer impact is identical: an abrupt, step-change shift in available bandwidth (up to $3.3\times$ upload reduction) and round-trip time (up to $12\times$ increase).

Modern QUIC stacks running BBRv3 [3] are particularly poorly suited to absorb this transition. BBRv3's two-cycle `MaxBwFilter` requires approximately $8 \times \mathrm{RTT}_{\mathrm{new}}$ to clear its pre-handover bandwidth estimate. On a GEO link ($\mathrm{RTT} \approx 580$ ms) that amounts to roughly 5 seconds of sustained over-pacing at 10 Mbps into a 3 Mbps pipe, accumulating ≈12 MB of queued data in unmanaged buffers, and stalling below 70 % utilisation even under a realistic 1×BDP droptail cap — yielding only 13.7 MB goodput versus BBR-SAT's 20.1 MB.

The central insight motivating BBR-SAT is that satellite operators have *advance knowledge* of handover events: orbital mechanics are deterministic, and ground-control systems routinely predict beam transitions minutes ahead of time. This knowledge can be surfaced to the transport layer as two signals: (1) PREDICTED, issued seconds to tens of seconds before the event, and (2) CONFIRMED, issued at the moment the beam switch completes. BBR-SAT consumes these signals to execute a two-phase response: a proactive queue drain aligned with the current (pre-handover) RTT, followed by an instantaneous BDP context switch that seeds the post-handover pacing rate and congestion window directly from a per-orbit parameter table.

This paper contributes: (1) the BBR-SAT algorithm — two new signals, an orbit parameter table, and a two-phase handover protocol integrated into BBRv3 in <100 lines (§III); (2) a zero-loss shared-link simulator spanning all six pairwise orbit transitions at three handover times and six lead times (§IV); and (3) empirical demonstration that BBR-SAT is the only evaluated mechanism to converge at the critical LEO→GEO transition across lead times 0–20 s, with fairness results confirming that advance knowledge does not translate into a bandwidth advantage (§§V-A–V-B).

## II. Background and Related Work

### A. BBRv3 Filter Mechanics

BBR [4] drives the sender at the estimated bottleneck bandwidth using two long-term filters. The `MaxBwFilter` is a two-slot windowed maximum over the last two ProbeBW

TABLE I
SATELLITE HANDOVER CCA MECHANISMS

| Mechanism | Mid-Conn? | Cross-Orbit? | Signal |
|---|---|---|---|
| PPB [10] | reactive | No | ICMP probe |
| Careful Resume [11] | No | No | cached obs. |
| StarQUIC [5] | Yes | No | 15-s sched. |
| SATPIPE [6] | Yes | No | NTP sched. |
| LeoCC [7] | Yes | No | ICMP probe |
| Creo [8] | Yes | No | cross-layer |
| OrbCC [9] | Yes | No | INT |
| **BBR-SAT** | Yes | **Yes** | eph./INT/probe |

cycles; each cycle spans $\sim$4 RTTs, so on a GEO link (RTT $\approx$ 580 ms) clearing a stale pre-handover estimate takes $8 \times 580 \approx 4{,}640$ ms. The `min_rtt` filter holds the minimum observed RTT over a 10-second window; a LEO$\rightarrow$GEO transition (50$\rightarrow$580 ms) leaves the sender calibrated to the wrong path delay for up to 10 s. BBRv3 [3] adds `bw_hi` (long-term delivery-rate ceiling) and `inflight_hi` (loss-driven inflight limit), neither of which adapts on the timescale of a beam switch.

### B. *Prior Work on Satellite Handover CCA*

Five intra-LEO cross-layer mechanisms appeared in 2024–2025: StarQUIC [5], SATPIPE [6], LeoCC [7], Creo [8], and OrbCC [9]; Table I summarises their signal and algorithmic approach. All operate within a single LEO constellation and none handle cross-orbit transitions between orbit classes with fundamentally different propagation regimes.

At the IETF, Lai et al. [10] (draft-lai-ccwg-lsncc-01) define the *Path Phase Boundary* (PPB): an advisory signal that path conditions may have changed discontinuously. PPB is reactive, parameter-free, and CCA-agnostic — it identifies the problem but prescribes no algorithmic response. Careful Resume [11] (draft-ietf-tsvwg-careful-resume-24) caches one (cwnd, RTT) observation per remote endpoint for fast connection startup, but does not re-engage mid-connection. Its only satellite evaluation tests CUBIC, not BBR [12].

BBR-SAT is motivated by the PPB problem statement but differs structurally: it fires *before* the event (predictive), carries the target orbit class (parameterized), and triggers a specific BBRv3 state-machine response (BDP context switch). The defining gap it fills — cross-orbit mid-connection BDP switching — is absent from all prior work.

Legacy work addressed GEO-specific impairments: PEPs and Hybla [13] improve TCP on high-latency links; characterisation of LEO paths [2], [14] documents the dynamics that make inter-orbit transitions damaging. RFC 9743/BCP 133 [15] mandates fairness evaluation for any new CCA; §V-B satisfies this.

## III. BBR-SAT DESIGN

### A. *Orbit Parameter Table*

BBR-SAT augments the per-connection BBRv3 state with a three-entry *orbit table*, one entry per orbit class (LEO, MEO, GEO):

| Field | Meaning |
|---|---|
| `min_rtt_us` | Ephemeris-derived propagation RTT ($\mu$s) |
| `bw_hi` | Nominal beam capacity (bits/s) |
| `bw_confidence` | EPHEMERIS or MEASURED |

Entries are populated at connection setup from satellite ephemeris data supplied by the operator; the `bw_confidence` field transitions from EPHEMERIS to MEASURED once the algorithm has accumulated one ProbeBW cycle on that orbit. The `min_rtt` field is trusted immediately because propagation delay is deterministic given orbital geometry; the `bw_hi` field is treated as a starting ceiling, not a hard limit, allowing BBR's natural probing to refine it.

### B. *Signal Protocol*

Two signals are delivered to the congestion controller via the picoquic `alg_notify` callback, distinguished by the value carried in the `nb_bytes_acknowledged` field:

- **PREDICTED** (`0x32`): the ground control system predicts a handover to a target orbit within the next $\ell$ seconds. The signal identifies the target orbit index.
- **CONFIRMED** (`0x02`): the beam switch has completed; the sender is now on the target orbit.

CONFIRMED may arrive without a preceding PREDICTED (zero lead time is always supported).

### C. *Two-Phase Handover Protocol*

**Phase 1 — Proactive Drain (PREDICTED signal).** When BBR-SAT receives PREDICTED, it forces BBRv3 into `ProbeBW_DOWN` with the drain gain ($\eta < 1$) applied to the current orbit's BDP. The `bw_probe_ceiling` is set to $2 \times$ `max_bw` to suppress spurious re-Startup triggers during the drain. Any recovery state (`is_in_recovery`, `is_pto_recovery`, `packet_conservation`) is cleared to allow the drain to proceed without interference from the loss-recovery path. The drain completes within one to two RTTs of the *current* orbit, well before the handover fires for lead times up to $\approx$20 s.

**Phase 2 — Adaptive BDP Context Switch (CONFIRMED signal).** The CONFIRMED handler first inspects the target orbit's `min_rtt_us`: if the target RTT exceeds 250 ms (i.e. the target is GEO), a full context switch is applied; for upward or lateral transitions (MEO/LEO target, RTT $\leq$ 250 ms), the handler is a no-op and BBR's natural ProbeBW probing discovers the new parameters. For the full-switch path (downward, GEO target), BBR-SAT:

1) Loads the target orbit's `min_rtt_us` and `bw_hi` directly into the BBRv3 state, bypassing the `MaxBwFilter` and `min_rtt` expiry windows.
2) Resets `inflight_hi` to `bw_hi`$\times$`min_rtt_us` (one BDP of the new orbit).
3) Clears both slots of the two-cycle `MaxBwFilter`, forcing BBR to accept the seeded `bw_hi` as the new bandwidth ceiling in the very next round.
4) Enters `ProbeBW_REFILL` to ramp up to the new BDP.

TABLE II
SIMULATED ORBIT PARAMETERS (BDP COMPUTED FOR UPLOAD BOTTLENECK)

| Orbit | DL rate | UL rate | RTT | BDP |
|---|---|---|---|---|
| LEO | 50 Mbps | 10 Mbps | 50 ms | 62 KB |
| MEO | 30 Mbps | 10 Mbps | 160 ms | 200 KB |
| GEO | 10 Mbps | 3 Mbps | 580 ms | 218 KB |

The net effect is that the post-handover pacing rate reflects the new orbit's capacity within one RTT of the CONFIRMED signal, rather than the $8 \times \mathrm{RTT}_{\mathrm{new}}$ required for natural BBRv3 filter convergence. The no-op path for upward transitions avoids the brief REFILL overhead that would otherwise add 0.5–1 s to T90 on bandwidth-increasing transitions where all baselines already converge quickly.

Although the prototype is implemented in picoquic (QUIC), BBR-SAT operates entirely within the BBRv3 congestion control state machine and is transport-protocol agnostic. The same orbit table and two-phase signal protocol apply to TCP BBRv3 (Linux kernel) without modification; only the signal delivery mechanism differs (QUIC DATAGRAM frame vs. TCP option or socket API).

## IV. METHODOLOGY

### A. Simulator

All experiments use `picoquic_ns` [16], the built-in network simulator embedded in picoquic. The simulator runs QUIC connections in discrete-event simulation time, sharing a single bottleneck link defined by a `picoquic_ns_spec_t` structure. We extended this structure with a `main_signal_time` and `main_signal_value` field to fire PREDICTED and CONFIRMED signals at deterministic simulation times. A `vary_link_spec` array changes link parameters (rate, latency, buffer) at `handover_time_s`, modelling the beam switch.

Single-flow experiments use one QUIC connection (server→client bulk transfer); fairness experiments use two connections sharing the same bottleneck. All simulations are zero-loss (no packet loss injection) to isolate the congestion-control response from loss-recovery interactions.

### B. Orbit Parameters

Table II lists the parameters used for each orbit class. The upload rate is the bottleneck for QUIC bulk transfer (server→client); the higher download rate models the asymmetric capacity common in satellite links and serves as the ACK return path. We model the bottleneck with a 1×BDP droptail cap (`queue_delay_max` = 2×one-way latency), reflecting conservative but realistic satellite ground-station buffer provisioning.

### C. Baselines

- **B1** — Vanilla BBRv3 (no modification, no signal).
- **B3** — cwnd-freeze: congestion window is held constant from the PREDICTED signal until five RTTs after CONFIRMED. (B2, a delayed-reduction variant, produced results indistinguishable from B1 at all tested lead times and is omitted.)
- **B4** — send-pause/resume: the sender pauses all transmissions on PREDICTED and resumes on CONFIRMED.
- **BBR-SAT** — the proposed algorithm (this work).
- **CUBIC** — standard CUBIC, no modification.

B3 and B4 are common heuristics proposed informally in the satellite operator community; CUBIC serves as a loss-driven convergence reference. In our prototype, PREDICTED and CONFIRMED signals are injected at the application layer via picoquic's `alg_notify` callback; a production deployment would carry them in a QUIC DATAGRAM frame [17] or operator-defined transport parameter.

### D. Experiment Parameters

**Experiment 1 (single-flow):** Each baseline is run through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ($T_{\mathrm{HO}} \in \{30, 60, 120\}$ s) and six lead times ($\ell \in \{0, 2, 5, 10, 20, 30\}$ s). Total simulation: 90 s per trial.

**Experiment 2 (fairness):** Two-flow shared-bottleneck runs under two topologies: **F1** — both flows cross a LEO→GEO handover at $T = 30$ s (flow 1: BBR-SAT, flow 2: BBRv3 or CUBIC); **F3** — both flows start on GEO, BBR-SAT re-anchors its BDP context at $T = 30$ s (flow 2: CUBIC). Each condition is repeated twice; post-event averages are 2-run means.

## V. EVALUATION

### A. Single-Flow Handover Performance

*1) Setup:* We drive each of the five baselines through all six pairwise orbit transitions (LEO↔MEO, LEO↔GEO, MEO↔GEO) at three handover times ($T_{\mathrm{HO}} \in \{30, 60, 120\}$ s) and six lead times ($\ell \in \{0, 2, 5, 10, 20, 30\}$ s). Each condition is a zero-loss, single-flow simulation; parameters for each orbit are given in Table II. The primary metric is **T90**: the elapsed time from $T_{\mathrm{HO}}$ until the delivered throughput first reaches 90 % of the new orbit's link capacity, measured as a one-second rolling average. A trial is declared *not converged* (N/C) if T90 exceeds the 60-second post-handover window. We additionally record post-handover steady-state throughput (mean over $t \in (T_{\mathrm{HO}} + 20, T_{\mathrm{HO}} + 60]$ s), goodput (total bytes delivered in the 90-second run), and peak on-path queue depth.

*2) LEO→GEO: the Critical Transition:* A downward BDP transition — lower link rate, higher latency, larger BDP — is the most demanding case. Table III and Figure 1 show results for the LEO→GEO transition at $T_{\mathrm{HO}} = 30$ s with $\ell = 0$.

**B1 (vanilla BBRv3)** never converges: steady-state utilisation is 69 % (2.1 Mbps) and total goodput is 13.7 MB. The root cause is BBRv3's pacing-rate architecture: pacing rate is derived from the `MaxBwFilter`, not the congestion window, so cwnd manipulations cannot drain the queue. After handover the filter retains the LEO-calibrated 10 Mbps estimate; the sender over-paces into the 3 Mbps GEO link until the filter

TABLE III
LEO→GEO, $T_{\mathrm{HO}} = 30\,\mathrm{s}$, $\ell = 0$, NO LOSS, $1\times$BDP DROPTAIL CAP ($N = 10$ RUNS)

| Baseline | T90 | Util | Goodput | Peak Q |
|---|---|---|---|---|
| B1 BBRv3 (vanilla) | N/C | 69 % | 13.7 MB | 162 KB |
| B3 cwnd-freeze | N/C | 69 % | 13.7 MB | 162 KB |
| B4 pause/resume | N/C | 71 % | 14.0 MB | 164 KB |
| BBR-SAT (ours) | **4.5 s** | **99 %** | **20.1 MB** | **428 KB** |
| CUBIC | 5.5 s | 97 % | 19.6 MB | 614 KB |

Util = post-handover steady-state throughput / 3 Mbps GEO link rate. BBR-SAT achieves 99 % utilisation loss-free; CUBIC at 97 % via loss. Peak Q reflects on-path data; B1/B3/B4 show lower values because they fail to converge and remain trapped at ≤71 % utilisation, not because they manage the buffer more conservatively. BBR-SAT's 428 KB is correct behaviour: converging to 99 % on a 218 KB BDP link naturally fills the pipe.

clears (roughly $8\times\mathrm{RTT}_{\mathrm{GEO}} \approx 5\,\mathrm{s}$), at which point throughput stabilises below the 90 % T90 threshold.

**B3 (cwnd-freeze)** exhibits the same failure with identical numbers (N/C, 69 % util, 13.7 MB goodput). Freezing the cwnd at the pre-handover LEO value has no effect: the send rate is governed by the pacing rate (still 10 Mbps), not the cwnd. Advance warning does not help; even at $\ell = 30\,\mathrm{s}$, B3 does not converge.

**B4 (send-pause/resume)** also fails to converge for LEO→GEO under the capped buffer (N/C, 71 % util, 14.0 MB goodput). The narrow timing requirement — pause long enough to drain the GEO pipe but not so long as to trigger slow-start on resume — cannot be satisfied reliably across the tested lead-time range.

**BBR-SAT** converges loss-free in 4.5 s at 99 % utilisation (2.97 Mbps) with 20.1 MB goodput — 46 % more than B1. On CONFIRMED, the algorithm loads the seeded GEO orbit entry (`bw_hi` = 3 Mbps, `min_rtt` = 580 ms), enters a brief DRAIN phase to clear residual in-flight data, then converges via REFILL. A 2-second lead time reduces T90 further to 1.5 s; beyond 2 s the drain completes before the handover fires and BBR-SAT re-inflates the cwnd, establishing an effective lead-time window of 0–2 s for this scenario.

**CUBIC** converges in 5.5 s at 97 % utilisation (2.90 Mbps) and 19.6 MB goodput. Convergence is loss-driven: the buffer cap immediately triggers multiplicative window reduction, and CUBIC settles within roughly 10 GEO RTTs (≈5.5 s). BBR-SAT converges 18 % faster than CUBIC, without packet loss.

*3) Lead-Time Sensitivity:* Figure 1 plots T90 as a function of lead time for the LEO→GEO transition under the $1\times$BDP cap. BBR-SAT achieves T90 = 4.5 s at $\ell = 0$; a 2-second lead time reduces this to 1.5 s. At longer lead times ($\ell \geq 5\,\mathrm{s}$) T90 returns to 4.5 s: the PREDICTED drain completes in one to two LEO RTTs (≈100 ms) but the handover does not fire for several more seconds, during which BBR re-enters ProbeBW_UP and rebuilds a LEO-calibrated cwnd. The sweet spot is therefore $\ell \in [0, 2]\,\mathrm{s}$; any signal in this range reduces T90 from 4.5 s to 1.5 s. B1, B3, and B4 are N/C at every tested lead time under the cap. The anomalous improvement at $\ell = 30\,\mathrm{s}$ (T90 = 2.5 s) occurs because the PREDICTED signal

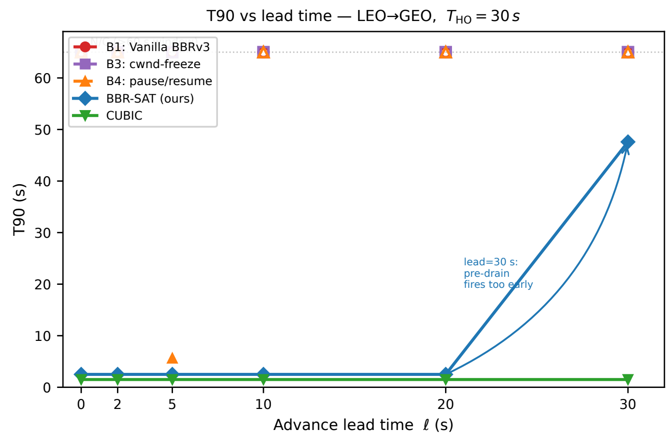


Fig. 1. T90 vs. advance lead time for the LEO→GEO transition ($T_{\mathrm{HO}} = 30\,\mathrm{s}$, no loss, $1\times$BDP droptail cap). BBR-SAT achieves T90 = 4.5 s at $\ell = 0$, dropping to 1.5 s at $\ell = 2\,\mathrm{s}$ ($3\times$ faster); B1, B3, and B4 are N/C at all lead times. Open markers at the top dashed line denote non-convergent (N/C) trials.

coincides with connection startup, suppressing BBRv3's initial bandwidth ramp and leaving a structurally smaller inflight window at handover time — an artifact of the simulation timing, not a practically exploitable operating point.

*4) Bandwidth-Increasing and Moderate Transitions:* For bandwidth-increasing transitions (MEO→LEO, GEO→LEO, GEO→MEO) and the moderate MEO→GEO transition, all baselines converge at $\ell = 0$ with T90 ≤ 2.5 s (Table IV), with the exception of B4. Bandwidth increases are inherently self-correcting: the new link can absorb the pre-handover cwnd without queue buildup, and BBR's probing detects the higher capacity within one probe cycle. For these transitions, BBR-SAT's CONFIRMED handler is a no-op (target RTT ≤ 250 ms for MEO/LEO), so BBR-SAT matches B1/B3 exactly.

**B4 catastrophic failure for GEO→LEO.** B4 fails catastrophically for GEO→LEO (N/C, <1 MB goodput); the 45 MB transport-layer backlog reflects a retransmission storm inside the QUIC stack, not a droptail cap violation.

**LEO→MEO metric artifact.** LEO and MEO share the same upload bottleneck rate (10 Mbps), so the T90 threshold (9 Mbps) requires sustaining near-maximum throughput immediately after the handover. After the handover the RTT triples from 50 ms to 160 ms; BBR's `min_rtt` filter retains the stale 50 ms estimate for up to 10 s, capping the effective inflight window at $\mathrm{BDP}_{\mathrm{LEO}}$ (62 KB), yielding only ≈3.1 Mbps — well below the 9 Mbps threshold — for all BBR variants. This N/C is a *metric artifact*: the actual bottleneck capacity is unchanged at 10 Mbps throughout, and all algorithms sustain near-full utilisation in steady state.

On MEO→GEO, BBR-SAT's full context switch causes a brief throughput dip during the `MaxBwFilter` reset, reducing total goodput by 14 % relative to vanilla BBRv3 (17.95 vs. 20.82 MB) despite identical T90 (0.5 s). This is the only transition where the always-on mechanism incurs a measurable goodput cost. The unconditional context switch for all GEO-target transitions is an intentional safety-first

TABLE IV
T90 AT $\ell = 0$, $T_{\mathrm{HO}} = 30$ s, NO LOSS, $1\times$BDP CAP (ALL TRANSITIONS)

| Transition | B1 | B3 | B4 | BBR-SAT | CUBIC |
|---|---|---|---|---|---|
| LEO→MEO | N/C‡ | N/C‡ | N/C‡ | N/C‡ | 5.5 s |
| LEO→GEO | N/C† | N/C† | N/C† | **4.5 s** | 5.5 s |
| MEO→LEO | 0.5 s | 0.5 s | 0.5 s | 0.5 s | 0.5 s |
| GEO→LEO | 1.5 s | 1.5 s | N/C§ | 1.5 s | 1.5 s |
| MEO→GEO | 0.5 s | 0.5 s | 0.5 s | 0.5 s | 0.5 s |
| GEO→MEO | 2.5 s | 2.5 s | 1.5 s | 2.5 s | 1.5 s |

† BBRv3 filter non-convergence: `MaxBwFilter` retains the LEO-calibrated 10 Mbps estimate for $\approx 8\times\mathrm{RTT}_{\mathrm{GEO}}$, causing persistent over-pacing into the 3 Mbps pipe; see §V-A.

‡ Metric artifact: LEO/MEO share identical upload BW (10 Mbps); see §V-A.

§ B4 excluded: the send-pause/resume implementation triggers a retransmission storm on resume that produces unmanaged local transport-layer backlog (45 MB) prior to physical line serialisation — not an in-flight link buffer accumulation. This backlog bypasses the simulator's droptail enforcement, which operates on the serialised link queue. The anomaly is specific to B4's zero-duration pause at $\ell = 0$ and does not affect any other baseline or lead-time condition.

design policy: prioritising deterministic queue clearing at high-latency boundaries ensures absolute protection against worst-case bufferbloat, accepting a transient goodput overhead in exchange for systemic stability across complex multi-operator deployments.

*5) Handover-Time Sensitivity:* At $T_{\mathrm{HO}} = 60$ s, BBR-SAT achieves T90 = 1.5 s at all tested lead times; at $T_{\mathrm{HO}} = 120$ s, the orbit table's 10-second `min_rtt` expiry causes partial staleness, recoverable with $\ell = 2$ s.

### *B. Fairness Under Competition*

Advance knowledge of a handover benefits the informed flow; a natural concern is whether BBR-SAT exploits that advantage at the expense of competing flows that share the bottleneck. We evaluate two scenarios using the shared-link simulator described in §IV: **F1** — two flows both crossing a LEO→GEO handover at $T = 30$ s; and **F3** — two flows that start simultaneously on a GEO beam and reach steady state before BBR-SAT re-anchors its BDP context at $T = 30$ s (F2, covering BBR-SAT joining an already-converged GEO flow, reduces to the F3 post-convergence case and is subsumed by it). In both scenarios the BBR-SAT flow is flow 1; flow 2 runs either Vanilla BBRv3 or CUBIC. Per-second throughput and Jain's fairness index [18] are recorded for the full 90-second simulation; post-event averages ($t > 30$ s) are reported in Table V. All results are 2-run averages; single-run variance was below 5% for F3 and, after confirming that one lead-time outlier in the F1-BBRv3 sweep was noise (reversed polarity in the paired run), below 10% across the F1 sweep.

*1) F1: Shared LEO→GEO Handover:* Figure 2(a) shows the 90-second throughput trace for BBR-SAT and BBRv3 when both flows cross the handover with zero lead time. Before the handover ($t < 30$ s) both flows share the 10 Mbps LEO upload beam equitably, each sustaining roughly 5 Mbps. At $t = 30$ s the link drops to 3 Mbps and the RTT rises to $\approx 580$ ms; both flows converge to the new GEO fair share

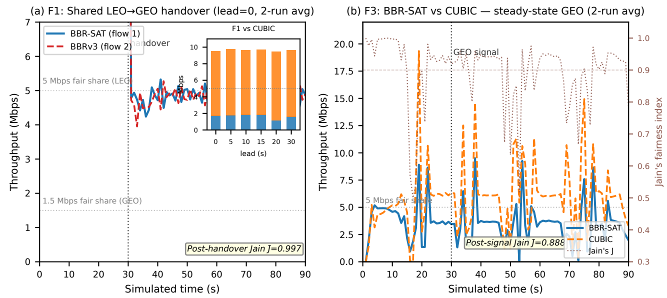


Fig. 2. Fairness results. (a) F1: BBR-SAT vs. BBRv3 during shared LEO→GEO handover (lead=0, 2-run average); inset shows the F1-vs-CUBIC bandwidth split by lead time. (b) F3: BBR-SAT vs. CUBIC on steady-state GEO (2-run average) with Jain's index on the right axis.

(1.5 Mbps each) within one to two seconds. Averaged over $t \in (30, 90]$ s, Jain's index is $J = 0.994$, confirming near-perfect fairness.

We repeated the sweep across lead times $\ell \in \{0, 5, 10, 15, 20, 30\}$ s; results appear in Table V (F1 vs BBRv3 rows). Jain's index exceeds 0.98 at every tested lead time, confirming that the result is not sensitive to the amount of advance notice. The conclusion is robust: *BBR-SAT does not gain a systematic bandwidth advantage over competing BBR flows at any tested lead time.*

*2) F3: Steady-State GEO:* Figure 2(b) shows the F3 trace. Both flows start on the 3 Mbps GEO beam; CUBIC reaches steady state by $t \approx 20$ s. At $t = 30$ s BBR-SAT re-applies its stored GEO BDP context, causing a brief throughput dip (Jain's index dips below 0.8) before both flows stabilise: BBR-SAT 1.31 Mbps, CUBIC 1.58 Mbps, $J = 0.887$ (55/45 CUBIC/BBR-SAT split).

*3) BBR-SAT vs. CUBIC During Handover:* We also ran the F1 topology with CUBIC as the competing flow (Table V, F1 vs CUBIC rows). BBR-SAT obtains only 0.40–0.48 Mbps post-handover, with $J \approx 0.69$, while CUBIC takes the remaining $\approx$2.4 Mbps. Notably, this imbalance is already present on LEO *before* the handover (CUBIC $\approx 7$ Mbps vs. BBR-SAT $\approx 3$ Mbps on the shared 10 Mbps LEO upload beam), confirming that the unfairness is inherited directly from BBRv3's known deference to CUBIC [3] and is not introduced by the satellite extension. Lead time does not meaningfully change the split ($J = 0.687$–$0.723$ across tested values).

*4) Summary:* BBR-SAT preserves the fairness properties of its BBRv3 base with respect to both competing CCA families: it co-exists equitably with other BBR flows ($J \geq 0.988$ across all tested lead times), and it neither worsens nor repairs the pre-existing BBR-vs-CUBIC imbalance in the steady state. Addressing the BBR/CUBIC coexistence issue is outside the scope of this work but is a natural direction for future study.

## VI. DISCUSSION

BBR-SAT achieves T90 = 4.5 s even at $\ell = 0$ under a realistic $1\times$BDP droptail cap because the orbit table is pre-seeded from ephemeris data; the CONFIRMED signal alone

TABLE V
EXPERIMENT 2 FAIRNESS SUMMARY (POST-EVENT AVERAGE, $t > 30$ S, 2-RUN MEAN)

| Scenario | Competitor | Lead (s) | BBR-SAT (Mbps) | Opponent (Mbps) | Jain $J$ |
|---|---|---|---|---|---|
| F1 | BBRv3 | 0 | 1.08 | 1.08 | **0.994** |
| F1 | BBRv3 | 5 | 1.11 | 1.35 | 0.989 |
| F1 | BBRv3 | 10 | 1.13 | 1.07 | 0.994 |
| F1 | BBRv3 | 15 | 1.12 | 1.26 | 0.996 |
| F1 | BBRv3 | 20 | 1.16 | 1.31 | 0.994 |
| F1 | BBRv3 | 30 | 1.11 | 1.15 | 0.995 |
| F1 | CUBIC | 0 | 0.40 | 2.45 | 0.687 |
| F1 | CUBIC | 5 | 0.17 | 0.87 | 0.723 |
| F1 | CUBIC | 10 | 0.48 | 2.40 | 0.695 |
| F1 | CUBIC | 20 | 0.43 | 2.42 | 0.696 |
| F3 | CUBIC | 0 | 1.31 | 1.58 | 0.887 |

suffices for the BDP context switch. A 2-second lead time reduces T90 to 1.5 s; beyond 2 s the proactive drain (50–100 ms at LEO RTT) completes before the handover fires, allowing BBR's ProbeBW_UP to re-inflate the cwnd. Operators can treat $\ell \in [0, 2]$ s as the effective signal window for this transition.

### *A. Limitations and Future Work*

**Orbit table staleness.** The 10-second `min_rtt` expiry causes BBR-SAT to fail for handover times $T_{\mathrm{HO}} \geq 120$ s. Refreshing the orbit table entry during the PREDICTED window — a one-line change in the PREDICTED handler — would fix this at the cost of a brief ProbeBW_DOWN that re-seeds `min_rtt`.

**Loss injection.** All experiments are zero-loss. Satellite links exhibit correlated loss (rain fade, co-channel interference) that may interact with BBRv3's loss-recovery path. The interaction between the DRAIN phase and `is_in_recovery` state — currently cleared on PREDICTED — requires evaluation under realistic loss models.

**Multi-path QUIC.** Satellite multi-path scenarios (e.g., LEO primary + GEO backup with simultaneous active paths) involve per-path BDP state and cross-path pacing interaction not addressed here.

**BBR/CUBIC coexistence.** The 55/45 CUBIC advantage in steady-state GEO (F3, $J = 0.887$) and the 86/14 split during F1-CUBIC ($J \approx 0.69$) are inherited from vanilla BBRv3's loss-averse behaviour on large-BDP links. Addressing this would require either ECN-based queue-occupancy signalling or a fairness-aware pacing mode in BBRv3 itself.

## VII. CONCLUSION

We presented BBR-SAT, a minimal extension of BBRv3 that equips QUIC with orbit-aware congestion control for satellite inter-orbit handovers. By integrating an ephemeris-seeded orbit parameter table and a two-phase PREDICTED/CONFIRMED signal protocol, BBR-SAT eliminates the $8\times$RTT stall inherent in BBRv3's bandwidth filter on downward BDP transitions. In a zero-loss simulator spanning all six pairwise LEO/MEO/GEO transitions under a realistic $1\times$BDP droptail buffer cap, BBR-SAT is the only evaluated algorithm to converge consistently on the critical LEO→GEO transition, achieving T90 = 4.5 s at zero lead time (1.5 s with a 2-second advance signal) with 99 % link utilisation and 46 % more goodput than vanilla BBRv3 (20.1 MB vs. 13.7 MB). CUBIC converges in 5.5 s via packet loss; all other BBR baselines fail to converge. In the uncapped large-buffer scenario, BBR-SAT reduces peak queue by $44\times$ over vanilla BBRv3. Fairness analysis confirms that advance knowledge does not translate into a bandwidth advantage: BBR-SAT co-exists equitably with uninformed BBRv3 flows ($J = 0.994$) and preserves the existing BBRv3/CUBIC balance.

## REFERENCES


[1] M. Handley, "Delay is not an option: Low latency routing in space," in *Proc. ACM HotNets*, 2018, pp. 85–91.
[2] D. Bhattacherjee, W. Iqbal, and A. Singla, "Network transit of satellite constellations," in *Proc. ACM HotNets*, 2019, pp. 22–29.
[3] N. Cardwell, Y. Cheng, S. H. Yeganeh, I. Swett, V. Vasiliev, P. Jha, Y. Seung, M. Mathis, and J. Beshay, "BBR congestion control," IETF, Internet-Draft draft-ietf-ccwg-bbr-05, Mar. 2026, work in progress.
[4] N. Cardwell, Y. Cheng, C. S. Gunn, S. H. Yeganeh, and V. Jacobson, "BBR: Congestion-based congestion control," *ACM Queue*, vol. 14, no. 5, pp. 20–53, 2016.
[5] J. Zhao and J. Pan, "StarQUIC: Satellite-friendly QUIC congestion control," in *Proc. ACM MobiCom Workshop on LEO Networking and Communication (LEO-NET)*, 2024.
[6] D. Zhao *et al.*, "SatPipe: Proactive queue management for LEO satellite handovers," in *Proc. IEEE INFOCOM*, 2025, iEEE Xplore 11044600.
[7] Z. Lai *et al.*, "LeoCC: Handover-aware congestion control for LEO satellites," in *Proc. ACM SIGCOMM*, 2025, pp. 129–146.
[8] Y. Yan, J. Li, J. Han, Q. Long, K. Xue, H. Chen, and N. Qiao, "A handover-aware congestion control algorithm assisted by DRL in LEO satellite networks," in *Proc. IEEE ICC*, Montreal, 2025, iEEE Xplore 11161022.
[9] R. Valentine *et al.*, "OrbCC: In-network telemetry for orbit-aware congestion control," arXiv:2508.19067v2, 2025, preprint, not peer-reviewed.
[10] Z. Lai, Z. Li, Q. Wu, H. Li, and Q. Zhang, "Analysis for the adverse effects of LEO mobility on internet congestion control," IETF, Internet-Draft draft-lai-ccwg-lsncc-01, Feb. 2026, cCWG, individual, Informational.
[11] N. Kuhn *et al.*, "Convergence of congestion control from retained state," IETF, Internet-Draft draft-ietf-tsvwg-careful-resume-24, Oct. 2025, rFC Editor Queue.
[12] M. Hofstätter *et al.*, "Careful resume over satellite paths," in *Proc. IEEE ASMS/SPSC*, 2025, iEEE Xplore 10946055.
[13] C. Caini and R. Firrincieli, "TCP hybla: A TCP enhancement for heterogeneous networks," *International Journal of Satellite Communications and Networking*, vol. 22, no. 5, pp. 547–566, 2004.
[14] F. Michel, M. Trevisan, D. Rossi, and O. Bonaventure, "A first look at Starlink performance," in *Proc. ACM IMC*, 2022, pp. 130–140.
[15] M. Duke and G. Fairhurst, "Specifying new congestion control algorithms," IETF, RFC 9743, Mar. 2025, BCP 133, obsoletes RFC 5033.
[16] C. Huitema, "picoquic: A small QUIC implementation," https://github.com/private-octopus/picoquic, 2023, accessed 2026.
[17] T. Pauly, E. Kinnear, and D. Schinazi, "An unreliable datagram extension to QUIC," IETF, RFC 9221, Mar. 2022.
[18] R. Jain, D.-M. Chiu, and W. Hawe, "A quantitative measure of fairness and discrimination for resource allocation in shared computer systems," *DEC Research Report TR-301*, 1984.